

# Pengyi Peng

Research CV | AI for Finance | Research Automation | Data Intelligence

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Mathematics and complex-systems graduate developing research workflows at the intersection of AI infrastructure, quantitative finance, and real-world financial decision support. Interested in RA/PhD collaboration across the Hong Kong-Shenzhen research ecosystem, with a focus on reliable, human-in-the-loop systems that turn heterogeneous information into reproducible research.

## EDUCATION

King's College London, University of London Sep 2022 - Nov 2024

MSc, Complex Systems Modelling, Department of Mathematics

- Trained in stochastic processes, numerical methods, mathematical modelling, complex systems, and scientific computing.
- Dissertation research on quantum entanglement and strong-disorder renormalization group methods in one-dimensional disordered systems, using numerical simulation and research writing.

University of Reading / Nanjing University of Information Science and Technology Sep 2018 - Jul 2022

BSc, Mathematics and Applied Mathematics

- Foundation in real analysis, linear algebra, probability and statistics, differential equations, stochastic processes, and numerical analysis.

## SELECTED RESEARCH AND ENGINEERING PROJECTS

AI-Assisted Financial Research Workflow 2026

Personal Research Project

- Designed a retrieval-assisted research workflow for macro, rates, credit, foreign-exchange, bond-market, policy, and industry-chain information.
- Explored RAG, GraphRAG, and agent-style workflows for structured retrieval, event tracking, automated briefs, and analyst review.
- Framed financial research automation as a reliability problem involving evidence, temporal context, entity relationships, and human judgement.

Paper-To-Strategy Automation Study 2026

Quantitative Research Automation

- Studied the X2Strategy paper-to-strategy workflow: document parsing, layered strategy extraction, code generation, validation, backtesting, and diagnosis.
- Investigated implementation risks that can produce plausible but incorrect research results, including look-ahead timing, formula translation, and evaluation design.
- Developed an interest in reliable research agents that preserve assumptions, surface uncertainty, and support reproducible experiments.

MSc Dissertation: Quantum Entanglement And Complex Systems 2024

King's College London

- Applied mathematical modelling, literature review, and numerical simulation to study entanglement behaviour in disordered one-dimensional systems.
- Built research skills in problem formulation, computational experimentation, result interpretation, and technical communication.

## QUANTITATIVE AND FINANCIAL RESEARCH EXPERIENCE

WorldQuant 2024 - 2025

Quantitative Research

- Developed and evaluated alpha signals using price-volume features, time-series operators, cross-sectional ranking, and risk-filtered expressions.
- Applied neutralization, turnover control, Sharpe, fitness, robustness checks, and in-sample/out-of-sample review.
- Translated repeated research tasks into reusable ideas for hypothesis tracking, diagnostics, experiment review, and strategy cards.

Ping An Bank, Shenzhen Branch Jul 2025 - May 2026

Corporate Banking And Industry-Chain Research

- Supported industry-chain research, credit-preparation, transaction-banking, settlement, and cash-management workflows.
- Researched strategic industries including new-energy vehicles, advanced manufacturing, semiconductors, and internet ecosystems.
- Connected practical financial information needs with ideas for retrieval-based research support and structured decision workflows.

Positive Venture DAO / Shanghai Huiyan Private Fund 2023 - 2024

Quantitative Research Internships

- Studied crypto-market microstructure, liquidity, volatility, execution, factor research, and systematic strategy evaluation.

## CURRENT RESEARCH INTERESTS

- Knowledge-augmented and agentic systems for research automation.
- AI for finance, financial intelligence, and analyst-in-the-loop decision support.
- Reliable paper-to-code and paper-to-strategy pipelines with explicit validation and uncertainty handling.
- Complex systems, heterogeneous information, and quantitative modelling.

## TECHNICAL SKILLS

- Programming And Data: Python, pandas, NumPy, matplotlib, scikit-learn, Jupyter, Git/GitHub, Mathematica, Julia.
- AI And Research Infrastructure: RAG, GraphRAG, LightRAG study, LLM-agent workflows, document parsing, knowledge retrieval, automated research pipelines.
- Quantitative Research: Alpha research, factor construction, cross-sectional ranking, time-series operators, neutralization, signal diagnostics, backtest evaluation.
- Research: Literature review, numerical simulation, mathematical modelling, experiment design, research memo and technical writing.